



VELOCITY OF EXPECTATIONS



► For $f: X \rightarrow \mathbb{R}$ the velocity $\dot{\pi}_\beta$ measures how the expectation $\pi_\beta[f]$ change with

► **Proposition:** Assume π_β and f are sufficiently regular, we have

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\pi_{\beta}[f]=\pi_{\beta}[f\dot{\pi}_{\beta}]$$

$$\frac{\mathrm{d}}{\mathrm{d}\beta}\pi_{\beta}[f]=\frac{\mathrm{d}}{\mathrm{d}\beta}\int_{\mathcal{X}}f(x)\pi_{\beta}(x)\mathrm{d}x$$

$$= \int_x f(x) \frac{\mathrm{d}}{\mathrm{d}\beta} \pi_\beta(x) \mathrm{d}x$$

$$= \int_X f(x) \dot{\pi}_\beta(x) \pi_\beta(x) dx$$

$$= \pi_{\beta} [f \pi_{\beta}]$$

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- ▶ **Proof:** Assume π_β and f are sufficiently regular, we have

$$\begin{aligned} \frac{d}{d\beta} \pi_\beta[f] &= \frac{d}{d\beta} \int_{\mathbb{X}} f(x) \pi_\beta(x) dx \\ &= \int_{\mathbb{X}} f(x) \frac{d}{d\beta} \pi_\beta(x) dx \\ &= \int_{\mathbb{X}} f(x) \dot{\pi}_\beta(x) \pi_\beta(x) dx \\ &= \pi_\beta[f \dot{\pi}_\beta] \end{aligned}$$

VELCOCITY OF ANNEALING PATHS